

MTH

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe Grundfos MTH.

Sections 1-3 give the information necessary to be able to unpack, install and start up the product in a safe way.

Sections 4-8 give important information about the product, as well as information on service, fault finding and disposal of the product.

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1. General information

1.1 Hazard statements

The symbols and hazard statements below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.

DANGER



Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

WARNING



Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

CAUTION



Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The hazard statements are structured in the following way:

SIGNAL WORD



Description of hazard

Consequence of ignoring the warning.
- Action to avoid the hazard.



Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1.2 Notes

The symbols and notes below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



Observe these instructions for explosion-proof products.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

2. Installing the product

2.1 Mechanical installation

2.2 Tank mounting



CAUTION

Hot or cold surface

Minor or moderate personal injury
- Make sure that no one can accidentally come into contact with hot or cold surfaces.



The pump is designed for tank mounting in vertical position. The pump is positioned in a hole cut into the cover of the tank (upper side) and is secured to the tank by four hexagon head screws through the holes in the mounting flange. We recommend that you fit a sealing gasket between the pump flange and tank.



MTH pumps must only be mounted vertically.

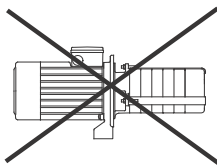
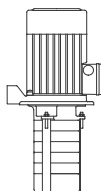


Fig. 1 Pump mounting position

2.2.1 Filtration requirements

A filtration system must be installed in the tank system to avoid that impurities in the liquid block the pump.

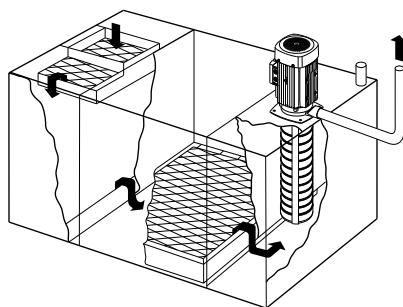


Fig. 2 Example of a filtration system

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TM07 0452 5117

2.2.2 Bypass pipe



To reduce the risk of water hammer damaging the pump, we recommend that you install a bypass line.

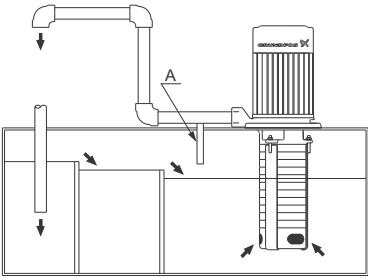


Fig. 3 Example of a bypass pipe (A)

2.2.3 Installation dimensions

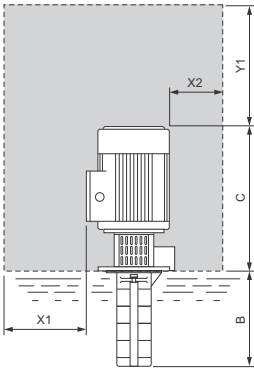


Fig. 4 Installation dimensions

Measurement	Dimension
B	Depending on pump type.
C	Depending on motor type.
X1	200 mm
X2	150 mm
Y1	This measurement must be at least 30 mm longer than measurement B to allow sufficient space for removing the pump in case of service and maintenance.

Pump mounting flange dimensions

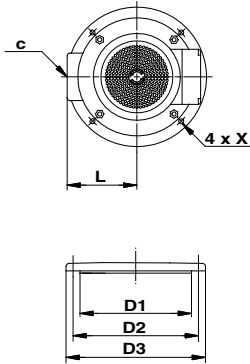


Fig. 5 Dimensional sketch

Pump type	D1	D2	D3	L	C	X
MTH 2, 4 and 8	140	160	180	121	Rp 3/4	Ø7.5
MTH 10 and 15	180	210	250	132	Rp/G 1 1/4	Ø9

2.3 Inlet conditions

The bottom of the pump strainer must be at least 25 mm above the bottom of the tank. See fig. 6 or 7.

MTH pumps must be installed vertically, and the drain hole in the motor stool must have access to the tank.

The pumps are designed to provide full performance down to a level of A mm above the bottom of the strainer.

At a liquid level between A and B mm above the bottom of the strainer, the built-in priming screw will protect the pump against dry running.

Pump type	A [mm]	B [mm]
MTH 2, 4 and 8	37	25
MTH 10 and 15	40	28

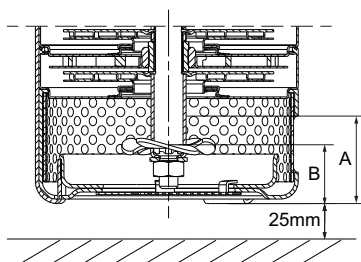


Fig. 6 MTH 2, 4 and 8

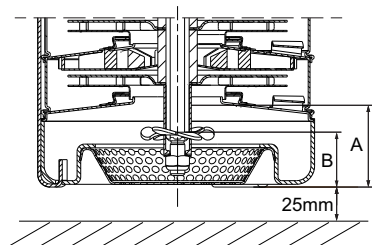


Fig. 7 MTH 10 and 15

2.4 Electrical connection

WARNING

Electric shock

Death or serious personal injury

- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on. Connect the pump to an external main switch close to the pump and to a motor-protective circuit breaker or a CUE frequency converter.

- Make sure you can lock the main switch in OFF position (isolated). Type and requirements as specified in EN 60204-1, 5.3.2.



Operating voltage and operating frequency are stated on the motor nameplate. Check that the motor is suitable for the power supply available at the installation site.

The voltage quality for MG motors, measured at the motor terminals, must be $\pm 10\%$ of the rated voltage during continuous operation (including variation in the supply voltage and losses in cables).

The electrical installation must be carried out by authorised persons in accordance with local regulations.

The wiring diagram is located in the terminal box cover.

The terminal box of single-speed motors normally contains six winding terminals and at least one earth terminal.

Single-phase motor

Connect single-phase motors to the mains in accordance with the instructions located in the terminal box cover.

Three-phase motor

Three-phase motors can be connected in star (Y) or delta (D) according to IEC 60034-8. See the wiring diagram in the terminal box cover.

Voltage and connection are stated on the nameplate.

Example: 220-240 V D / 380-415 V Y

- If the voltage supply is 220-240 V, the motor must be connected in delta.
- If the voltage supply is 380-415 V, the motor must be connected in star.

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2.4.1 Torque for terminal box cover screws

Torque: 1.6 - 2 Nm.

2.4.2 Motor protection

Single-phase motors

Single-phase motors are supplied with built-in thermal protection, according to IEC 60034-11, against thermal overload with both rapid and slow variation.

Three-phase motors

Three-phase motors must be protected by a motor-protective circuit breaker according to local regulations.

MG motors of 3.0 kW or up are supplied with thermal switches (PTC) as standard and are protected against thermal overload with both rapid and slow variation. The motor protection is stated on the nameplate.

2.4.3 Terminal box positions

The terminal box can be turned to three positions, in 90° steps, see fig. 8.

Proceed as follows:

1. Remove the four bolts securing the motor to the motor stool.
2. Turn the motor to the required position.
3. Replace and tighten the four bolts.

The electric motor must be connected to the supply as shown in the diagram inside the terminal box cover.

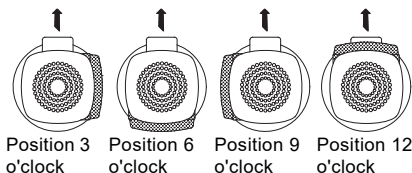


Fig. 8 Terminal box positions, top view

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2.4.4 Plug connections

The pumps can be fitted with a 10-pin multiplug connection, type Han® 10 ES.

The purpose of a multiplug connection is to make the electrical installation and the service of the pump easier.

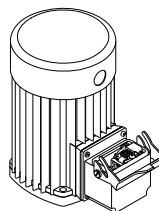


Fig. 9 Multiplug connection on a Grundfos motor

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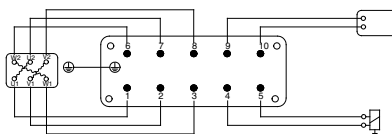


Fig. 10 Connections between motor terminals and plug terminals

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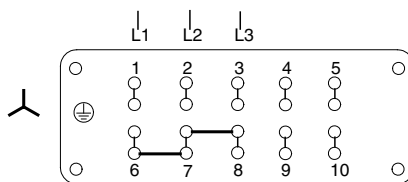


Fig. 11 Plug connections for star connection

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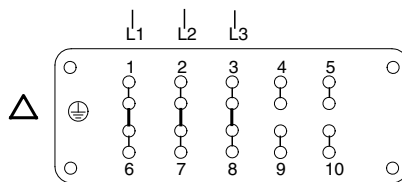


Fig. 12 Plug connections for delta connection, fishplates for connections are located in the plug

TM01 8704 0700

2.4.5 Frequency converter operation

Motors supplied by Grundfos

All three-phase motors supplied by Grundfos can be connected to a frequency converter.

Depending on the frequency converter type, this may cause increased acoustic noise from the motor. Furthermore, it may cause the motor to be exposed to detrimental voltage peaks.



Grundfos motors, types MG 71 and MG 80 as well as MG 90 (1.5 kW, 2-pole), for supply voltages up to and including 440 V (see motor nameplate), must be protected against voltage peaks higher than 650 V (peak value) between the supply terminals.

We recommend to protect all other motors against voltage peaks higher than 850 V.

The above disturbances, that is both increased acoustic noise and detrimental voltage peaks, can be eliminated by fitting an LC filter between the frequency converter and the motor.

For further information, please contact the frequency converter supplier or Grundfos.

3. Starting up the product

Before starting the pump, make sure:

- that all pipe connections are tight
- that the pump body is partly filled with liquid (partly submerged)
- that the strainer is not blocked by impurities.

Start the pump as follows:

1. Close the isolating valve on the outlet side of the pump.
2. See the correct direction of rotation of the pump on the motor fan cover. When seen from the top, the pump must rotate counterclockwise.
3. Start the pump and check the direction of rotation.
4. Slowly open the outlet valve until it is completely open.

The pump has now been vented and is ready for operation.

3.1 Operating the product

CAUTION



Fire and explosion

- Minor or moderate personal injury
- Do not use the pump for flammable, combustible or explosive liquids.

CAUTION



Hot liquid

- Minor or moderate personal injury
- Do not run the pump against a closed valve for more than approximately 5 minutes.



The pump is not allowed to run against a closed outlet valve for more than approximately 5 minutes as this will cause an increase in temperature and formation of steam in the pump which may cause damage to the pump.

4. Product introduction

4.1 Applications

Grundfos pumps, type MTH, are multistage centrifugal pumps designed for pumping liquids for machine tools, condensate transfer, liquid transfer in industrial washing machines and similar applications.

The pumps are designed for pumping liquids with a density and viscosity corresponding to those of water. The pumped liquid must not contain abrasive particles or fibres.

When pumping liquids with a density or viscosity higher than that of water, the motor size must be taken into consideration.

4.2 Identification

4.2.1 Type designation

The standard range consists of pumps with several impeller and chamber combinations. On request, other pump lengths can be supplied by fitting empty intermediate chambers in combination with standard impellers and chambers.

The type key on the pump nameplate indicates the number of chambers and impellers fitted to the pump.

4.2.2 Type key

Example	MTH 2 -6 /3 -A -W -A -A QQ V
Pump type	
Rated flow rate [m ³ /h]	
Number of chambers	
Number of impellers	
Pump version	
A Basic version	
C Inlet pipe	
X Special version	
Pipe connection	
W Internal thread	
WB NPT internal thread	
Materials	
A Basic version	
I Stainless steel pump head	
Shaft seal	
A O-ring seal with fixed seal driver	
B Bellows seal, rubber	
C O-ring seal with spring as seal driver	
R O-ring seal, type A, with reduced seal faces	
A Carbon, metal-impregnated	
B Carbon, resin-impregnated	
Q Silicon carbide	
U Tungsten carbide	
V Metal oxides, ceramic	
E EPDM	
V FKM	

5. Servicing the product

WARNING

Electric shock

Death or serious personal injury

- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on.
- Connect the pump to an external main switch close to the pump and to a motor-protective circuit breaker or a CUE frequency converter.
- Make sure you can lock the main switch in OFF position (isolated). Type and requirements as specified in EN 60204-1, 5.3.2.



CAUTION

Toxic liquids

Death or serious personal injury

- Wear personal protective equipment.



CAUTION

Hot or cold liquid

Minor or moderate personal injury

- Wear personal protective equipment.



CAUTION

Hot or cold surface

Minor or moderate personal injury

- Make sure that no one can accidentally come into contact with hot or cold surfaces.



5.1 Contaminated products

The product will be classified as contaminated if it has been used for a liquid which is injurious to health or toxic.

If you request Grundfos to service the product, contact Grundfos with details about the liquid before returning the product for service. Otherwise, Grundfos can refuse to accept the product for service.

Any application for service must include details about the liquid.

Clean the product in the best possible way before you return it.

5.2 Maintaining the product

Pumps installed in accordance with these instructions require very little maintenance.

Shaft seal

The mechanical shaft seal is self-adjusting and has wear-resistant seal rings which are lubricated and cooled by the pumped liquid.

Pump bearings

The pump bearings are also lubricated by the pumped liquid.

Motor bearings

The motor ball bearings are grease packed and sealed for life. No further lubrication is necessary.

Pumps from 4 kW and up have angular contact bearings.

5.3 Filters

Chip trays, filters, etc. must be cleaned at regular intervals to ensure a correct flow of liquid.

5.4 Periodic checks

At regular intervals, depending on the conditions and time of operation, the following checks must be made:

- Check the quantity of liquid and operating pressure.
- Check that there are no leaks.
- Check that the motor is not overheating.
- Check the tripping of the motor-protective circuit breaker.
- Check that all controls are operating satisfactorily.

If the above checks do not reveal any abnormal operating details, no further checks are necessary.

Should any faults be found, check the symptoms in section [6. Fault finding the product](#).

6. Fault finding the product

WARNING

Electric shock

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- Make sure you can lock the main switch in OFF position (isolated). Type and requirements as specified in EN 60204-1, 5.3.2.



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Toxic liquids

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CAUTION

Hot or cold liquid

- Minor or moderate personal injury
- Wear personal protective equipment.



CAUTION

Hot or cold surface

- Minor or moderate personal injury
- Make sure that no one can accidentally come into contact with hot or cold surfaces.



Fault	Cause	Remedy
1. The motor does not run when started.	a) Supply failure.	Connect the electricity supply.
	b) The fuses are blown.	Replace fuses.
	c) The motor-protective circuit breaker overload has tripped.	Reactivate the motor-protective circuit breaker.
	d) The main contacts in the motor-protective circuit breaker are not making contact or the coil is faulty.	Replace contacts or magnetic coil.
	e) The control circuit is defective.	Repair the control circuit.
	f) The motor is defective.	Replace the pump.
2. The motor-protective circuit breaker overload trips immediately when the supply is switched on.	a) One fuse is blown or the motor-protective circuit breaker has tripped.	Replace the fuse or reactivate the motor-protective circuit breaker.
	b) The contacts in the motor-protective circuit breaker overload are faulty.	Replace the motor-protective circuit breaker contacts.
	c) The cable connection is loose or faulty.	Fasten or replace the cable connection.
	d) The motor winding is defective.	Replace the pump.
	e) The pump is mechanically blocked.	Remove the mechanical blocking of the pump.
	f) The overload setting of the motor-protective circuit breaker is too low.	Set the motor-protective circuit breaker correctly.
3. The motor-protective circuit breaker overload trips occasionally.	a) The overload setting of the motor-protective circuit breaker is too low.	Set the motor-protective circuit breaker correctly.
	b) The voltage is low at peak times.	Check the electricity supply.
4. The motor-protective circuit breaker has not tripped but the pump does not run.	a) Check 1 a), b), d) and e).	

Fault	Cause	Remedy
5. The pump runs but gives no liquid or the pump capacity is not constant.	a) The pump strainer is partly blocked by impurities.	Clean the strainer.
	b) The liquid level in the tank is too low.	Increase the liquid level.
	c) The pump rotates in the wrong direction.	Change the direction of rotation of the motor.

7. Technical data

7.1 Operating conditions

Pump type	MTH
Minimum liquid temperature [°C]	-10
Maximum liquid temperature [°C]	+90
Maximum ambient temperature [°C]	+40
Maximum operating pressure [bar]	10
Enclosure class	IP54

7.1.1 Minimum flow rate

Pump type	Minimum flow rate [l/min]	
	At a liquid temperature of 80 °C	At a liquid temperature between 80 and 90 °C
MTH 2 and 4	3	8
MTH 8	13	33
MTH 10 and 15	20	50

7.1.2 Maximum number of starts

Maximum number of starts per hour: 100.

7.2 Sound pressure level

The sound pressure level of the pumps is lower than 70 dB(A).

8. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.



The crossed-out wheellie bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

See also end-of-life information at www.grundfos.com/product-recycling.

Declaration of conformity

GB: EC/EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the product MTH, to which the declaration below relates, is in conformity with the Council Directives listed below on the approximation of the laws of the EC/EU member states.

DE: EG-/EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt MTH, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-/EU-Mitgliedsstaaten übereinstimmt.

ES: Declaración de conformidad de la CE/UE

Grundfos declara, bajo su exclusiva responsabilidad, que el producto MTH al que hace referencia la siguiente declaración cumple lo establecido por las siguientes Directivas del Consejo sobre la aproximación de las legislaciones de los Estados miembros de la CE/UE.

FR: Déclaration de conformité CE/UE

Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit MTH, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des États membres CE/UE relatives aux normes énoncées ci-dessous.

IT: Dichiarazione di conformità CE/UE

Grundfos dichiara sotto la sua esclusiva responsabilità che il prodotto MTH, al quale si riferisce questa dichiarazione, è conforme alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE/UE.

PT: Declaração de conformidade CE/UE

A Grundfos declara sob sua única responsabilidade que o produto MTH, ao qual diz respeito a declaração abaixo, está em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE/UE.

RU: Декларация о соответствии нормам ЕЭС/ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделие MTH, к которому относится нижеприведённая декларация, соответствует нижеприведённым Директивам Совета Евросоюза о тождественности законов стран-членов ЕЭС/ЕС.

SK: EC/EU vyhlásenie o zhode

My, spoločnosť Grundfos, vyhlasujeme na svoju plnú zodpovednosť, že produkt MTH, na ktorý sa vyhlásenie uvedené nižšie vzťahuje, je v súlade s ustanoveniami nižšie uvedených smerníc Rady pre zblíženie právnych predpisov členských štátov EC/EU.

JP: EC/EU 適合宣言

Grundfos は、その責任の下に、MTH 製品が EC/EU 加盟諸国の法規に 関連する、以下の評議会指令に適合していることを宣言します。

TW: EC/EU 合格聲明

葛蘭富根據我們唯一的責任，茲聲明與以下聲明相關之 MTH 產品，符 合下列近似 EC/EU 會員國法律之議會指令。

CZ: Prohlášení o shodě EU

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobek MTH, na který se toto prohlášení vztahuje, je v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství.

DK: EF-/EU-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produktet MTH som erklæringen nedenfor omhandler, er i overensstemmelse med Rådets direktiver der er nævnt nedenfor, om indbyrdes tilnærmelse til EF-/EU-medlemsstaternes lovgivning.

FI: EY-/EU-vaatimustenmukaisuusvakuutus

Grundfos vakuuttaa omalla vastuullaan, että tuote MTH, jota tämä vakuutus koskee, on EY-/EU:n jäsenvaltioiden lainsäädännön lähentämiseen tähtäävien Euroopan neuvoston direktiivien vaatimusten mukainen seuraavasti.

GR: Δήλωση συμμόρφωσης ΕΚ/ΕΕ

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι το προϊόν MTH, στο οποίο αναφέρεται η παρακάτω δήλωση, συμμορφώνεται με τις παρακάτω Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΚ/ΕΕ.

NL: EG-/EU-conformiteitsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat product MTH, waarop de onderstaande verklaring betrekking heeft, in overeenstemming is met de onderstaande Richtlijnen van de Raad inzake de onderlinge aanpassing van de wetgeving van de EG-/EU-lidstaten.

RO: Declarația de conformitate CE/UE

Noi Grundfos declarăm pe propria răspundere că produsul MTH, la care se referă această declarație, este în conformitate cu Directivele de Consiliu specificate mai jos privind armonizarea legilor statelor membre CE/UE.

SE: EG-/EU-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkten MTH, som omfattas av nedanstående försäkran, är i överensstämmelse med de rättsdirektiv om inbördes närmande till EG-/EU-medlemsstaternas lagstiftning som listas nedan.

CN: 欧盟符合性声明

我们，格兰富，在我们的全权责任下声明，产品 MTH 系列，其制造和性能完全符合以下所列欧盟委员会指令。

KO: EC/EU 적합성 선언

Grundfos 는 아래의 선언과 관련된 MTH 제품이 EU 회원국 법률에 기 반하여 아래의 이사회 지침을 준수함을 단독 책임 하에 선언합니다.

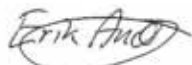
- Machinery Directive (2006/42/EC).
Standard used: EN 809:1998, A1:2009
- Ecodesign Directive (2009/125/EC).
Electric motors:
Commission Regulation No 640/2009.
Applies only to three-phase Grundfos motors marked IE2 or IE3.
See motor nameplate.
Standard used: EN 60034-30:2009.
Water pumps:
Commission Regulation No 547/2012.
Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.

This EC/EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 96440790).

Additional directives and standards effective from 22 July 2019:

- RoHS Directives: 2011/65/EU and 2015/863/EU
Standard used: EN 50581:2012

Bjerringbro, 17 December 2018



Erik Andersen
Senior Manager
GRUNDFOS Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark

Manufacturer and person empowered to sign the EC/EU declaration of conformity

RUS

МТН

Руководство по эксплуатации



Руководство по эксплуатации на данное изделие является составным и включает в себя несколько частей:

Часть 1: настоящее «Руководство по эксплуатации».

Часть 2: электронная часть «Паспорт. Руководство по монтажу и эксплуатации» размещенная на сайте компании Грундфос. Перейдите по ссылке, указанной в конце документа.

Часть 3: информация о сроке изготовления, размещенная на фирменной табличке изделия.

Сведения о сертификации:

Насосы типа МТН сертифицированы на соответствие требованиям Технических регламентов Таможенного союза: ТР ТС 004/2011 «О безопасности низковольтного оборудования»; ТР ТС 010/2011 «О безопасности машин и оборудования»; ТР ТС 020/2011 «Электромагнитная совместимость технических средств».

KAZ

МТН

Пайдалану бойынша нұсқаулық

Атаулы өнімге арналған пайдалану бойынша нұсқаулық құрамалы болып келеді және келесі бөлімдерден тұрады:

1 бөлім: атаулы «Пайдалану бойынша нұсқаулық»

2 бөлім: Грундфос компаниясының сайтында орналасқан электронды бөлім «Төлқұжат, Құрастыру және пайдалану бойынша нұсқаулық». Құжат соңында көрсетілген сілтеме арқылы өтіңіз.

3 бөлім: өнімнің фирмалық тақташасында орналасқан шығарылған уақыты жөніндегі мәлімет

Сертификаттау туралы ақпарат:

МТН типті сорғылары «Төмен вольтты жабдықтардың қауіпсіздігі туралы» (ТР ТС 004/2011), «Машиналар және жабдықтар қауіпсіздігі туралы» (ТР ТС 010/2011) «Техникалық заттардың электрлі магниттік сәйкестілігі» (ТР ТС 020/2011) Кеден Одағының техникалық регламенттерінің талаптарына сәйкес сертификатталды.

KG

МТН

Пайдалануу боюнча колдонмо

Аталган жабдууну пайдалануу боюнча колдонмо курамдык жана өзүнө бир нече бөлүкчөнү камтыйт:

1-Бөлүк: «Пайдалануу боюнча колдонмо»

2-Бөлүк: «Паспорт. Пайдалануу жана монтаж боюнча колдонмо» электрондук бөлүгү Грундфос компаниянын сайтында жайгашкан. Документтин аягында көрсөтүлгөн шилтемеге кайрылыңыз.

3-Бөлүк: жабдуунун фирмалык тактасында жайгашкан даярдоо мөөнөтү тууралуу маалымат.

Шайкештик жөнүндө декларация

МТН түрүндөгү соргучтар Бажы Биримдиктин Техникалык регламенттин талаптарына ылайыктуу тастыкталган:

ТР ТБ 004/2011 «Төмөн вольттук жабдуунун коопсуздугу жөнүндө»; ТР ТБ 010/2011 «Жабдуу жана машиналардын коопсуздугу жөнүндө»; ТР ТБ 020/2011 «Техникалык каражаттардын электрмагниттик шайкештиги».

ARM

МТН

Հահագործման ձեռնարկ

Տվյալ սարքավորման շահագործման ձեռնարկը բաղկացած է մի քանի մասերից.

Մաս 1. սույն «Հահագործման ձեռնարկ»:

Մաս 2. էլեկտրոնային մաս. այն է՝ «Անձնագիր: Մոնտաժման և

շահագործման ձեռնարկ» տեղադրված «Գրունդֆոս». Անցեք փաստաթղթի վերջում նշված հղումով.

Մաս 3. տեղեկություն արտադրման ամսաթվի վերաբերյալ՝ նշված սարքավորման պիտակի վրա:

Տեղեկություններ հավաստագրման մասին՝

MTH տիպի պոմպերը սերտիֆիկացված են համաձայն Մաքսային Միության տեխնիկական կանոնակարգի պահանջների՝ TP TC 004/2011 «Ցածրավոլտ սարքավորումների վերաբերյալ», TP TC 010/2011 «Մեքենաների և սարքավորումների անվտանգության վերաբերյալ» ; TP TC 020/2011 «Տեխնիկական միջոցների էլեկտրամագնիսական համատեղելիության վերաբերյալ»:



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